



SI 315: Models of Social Information Theory

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Logistics

- No AI-generated submissions to assignments
- Required attendance at both lectures and discussions
- Two exams, final is non-cumulative
- iClicker both participation and correctness
- Lowest two grades dropped from each of assignments, quizzes, discussion attendance, lecture attendance

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1 Introduction to Networks

1.1 Strong and Weak Ties

- Strong ties: frequent interaction, trust, emotional closeness
- Weak ties: infrequent interaction, limited intimacy
- Strong ties: overlapping social circles
- Weak ties: bridges between disconnected groups
- Weak ties: novel information and opportunities

1.2 Bridges and Local Bridges

- Bridge: an edge whose removal disconnects the network into separate components
- Local bridge: an edge whose endpoints do not have any direct ties to the same neighboring nodes
- Local bridge: removing it increases the distance between its endpoints, giving it a span of at least 3
- Span of a bridge: the distance between the endpoints after the edge is removed
- Bridges and local bridges: often connect otherwise distant or disconnected regions of the network

1.3 Clustering

- Clustering: connected neighbors also connected to one another
- Local clustering coefficient: connected neighbor pairs divided by possible neighbor pairs
- Formula: $C_i = \frac{2e_i}{k_i(k_i-1)}$
- e_i : edges among node i 's neighbors
- Network clustering coefficient: average local coefficient

1.4 Strong Triadic Closure Property

- Two strong ties sharing a node: likely third edge
- Strong ties around a node: mutually connected neighbors
- Open triad: two strong ties without a connecting edge
- Open triads: pressure toward closure
- Closed triads: higher clustering coefficients